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POPULATION VARIANCE

Average squared deviation from mean

Measures how far each data point is from the mean, squared and averaged.

$$\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n}$$

Units are squared, not original

RANGE

Simplest measure of dispersion

Difference between the highest and lowest values in a dataset, indicating spread.

$$Range = Max - Min$$

Ignores distribution shape

MODE (GROUPED)

Most frequent value in grouped data

Identifies the value occurring most often within the modal class of grouped data.

$$l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

f1=modal class frequency

MEDIAN (GROUPED)

Middle value in grouped data

Locates the central value of grouped frequency data using the cumulative frequency class.

$$l + \frac{(n/2 - cf)}{f} \times h$$

l=lower limit, cf=cum. freq.

ARITHMETIC MEAN

Sum of values divided by count

Average value of a dataset, calculated by dividing the sum of observations by number of items.

$$\bar{x} = \frac{\Sigma x}{n}$$

Sensitive to extreme outliers

DECK INDEX • PART 2

16: Poisson Dist.
17: Normal Dist.

18: Z-score

19: Std Error of Mean

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21: Chi-square Test

22: ANOVA

23: Laspeyres Index

24: Paasche Index

25: Fisher's Ideal Index

26: Pearson's Skewness

27: Kurtosis

28: Sampling Error

29: Law of Large Nos.

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KARL PEARSON'S R

Measures linear relationship strength

Quantifies the strength and direction of a linear relationship between two variables.

$$r = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\sqrt{\Sigma(x-\bar{x})^2\Sigma(y-\bar{y})^2}}$$

Ranges from -1 to +1

COEFFICIENT OF VARIATION

Relative measure of dispersion

Expresses standard deviation as a percentage of the mean, useful for comparisons.

$$CV = \frac{s}{\bar{x}} \times 100$$

Unitless, good for comparison

STANDARD DEVIATION

Square root of the variance

Most common measure of dispersion, expressed in the same units as original data.

$$\sigma = \sqrt{\sigma^2}$$

Lower value = less spread

CLASSICAL PROBABILITY

Ratio of favorable to total outcomes

Basic definition of probability assuming all outcomes are equally likely to occur.

$$P(E) = \frac{Favorable}{Total}$$

Value always between 0 and 1

REGRESSION (Y ON X)

Predicts Y from known X values

Linear equation used to estimate the dependent variable Y based on independent variable X.

$$Y = a + bX$$

b = regression coefficient/slope

SPEARMAN'S RANK CORR.

Correlation based on ranked data

Measures correlation using ranks instead of raw values, useful for ordinal data.

$$\rho = 1 - \frac{6\Sigma d^2}{n(n^2-1)}$$

d = difference in ranks

ADDITION
THEOREM

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MULTIPLICATION
THM

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BAYES'
THEOREM

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BINOMIAL
DIST.

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POISSON
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NORMAL
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Z-SCORE

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ANOVA

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LASPEYRES
INDEX

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PAASCHE
INDEX

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FISHER'S
IDEAL
INDEX

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PEARSON'S
SKEWNESS

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KURTOSIS

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SAMPLING
ERROR

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LAW
OF
LARGE
NOS.

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NORMAL DISTRIBUTION The bell-shaped continuous curve Symmetric continuous probability distribution defined completely by its mean and standard deviation. $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$Mnemonic: mean=median=mode	POISSON DISTRIBUTION Models rare events over an interval Discrete distribution for count of events occurring in a fixed time or space interval. $P(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}$$\lambda$ = mean number of events	BINOMIAL DISTRIBUTION Probability of r successes in n trials Discrete distribution modeling number of successes in fixed independent Bernoulli trials. $P(X = r) = {}^nC_rp^rq^{n-r}$p=success, q=failure prob.	BAYES' THEOREM Revises probability with new evidence Updates the probability of event A given evidence B has already occurred. $P(A B) = \frac{P(B A)P(A)}{P(B)}$By Thomas Bayes, 18th century	MULTIPLICATION THEOREM Joint probability of independent events Gives the probability of two independent events A and B both occurring together. $P(A \cap B) = P(A) \times P(B)$Only valid if A, B independent	ADDITION THEOREM Probability of A or B occurring Calculates the probability that at least one of two events A or B will occur. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$Subtract overlap to avoid double count
LASPEYRES PRICE INDEX Base-year quantities as weights Price index using base-year quantities as fixed weights to measure price changes over time. Named after Etienne Laspeyres Tends to overstate inflation	ANOVA Compares means across multiple groups Analysis of Variance tests whether variance between group means exceeds variance within groups. Developed by Ronald Fisher Uses F-statistic for testing	CHI-SQUARE TEST Tests observed vs expected frequencies Compares observed categorical data frequencies against expected frequencies to test independence. $\chi^2 = \sum \frac{(O-E)^2}{E}$O=observed, E=expected	ONE-SAMPLE T-TEST Tests sample mean vs population Checks whether a sample mean differs significantly from a known or hypothesized population mean. $t = \frac{\bar{x}-\mu}{s/\sqrt{n}}$Used when population σ unknown	STANDARD ERROR Variability of the sample mean Estimates how much a sample mean would vary from the true population mean. $SE = \frac{\sigma}{\sqrt{n}}$Shrinks as sample size grows	Z-SCORE Standardized distance from the mean Number of standard deviations a data point lies away from the population mean. $z = \frac{x-\mu}{\sigma}$Used to compare different scales
LAW OF LARGE NUMBERS Sample mean converges to true mean As sample size increases, the sample average approaches the expected population value. Proposed by Jacob Bernoulli Foundation of statistical inference	SAMPLING ERROR Gap between sample and population Deviation between a statistic calculated from a sample and the true population parameter. Decreases with larger sample size Unavoidable in sample-based study	KURTOSIS Measures peakedness of distribution Describes tail heaviness and peak sharpness of a distribution relative to normal curve. Types: leptokurtic, mesokurtic, platykurtic Mnemonic: Lepto = tall/thin peak	PEARSON'S SKEWNESS Measures asymmetry of distribution Indicates the direction and degree of asymmetry in a frequency distribution's shape. $Sk = \frac{3(Mean-Median)}{SD}$Zero means symmetric distribution	FISHER'S IDEAL INDEX Geometric mean of two indices Combines Laspeyres and Paasche indices via geometric mean to correct their individual biases. $F = \sqrt{L \times P}$Satisfies time reversal test	PAASCHE PRICE INDEX Current-year quantities as weights Price index that uses current-year quantities as weights, updating basket composition yearly. Named after Hermann Paasche Tends to understate inflation